The Thread

The current picture of the Relational Geometry Program

Le Matt Ansatz Di Ego · October 24, 2026

English translation of `HILO_ONTOLOGICO_INTEGRADO_2026-10-24.md`. The Spanish text is the authoritative version.

> What is real is what can be described. This text follows a single thread, from language to the branches, and at each step it says what is proved, what is interpretation, and what is still open.

About this translation

Spanish remains the reference. If the two versions ever disagree, the Spanish text governs. Folder and file names are left in Spanish, because that is what they are called in the package. The author's own passages, such as the image of the fruit and the mushroom in Part I, are translated as faithfully as possible and not rewritten.

Key terms. The program's vocabulary is fixed by its semantic crystal, a glossary of 100 terms with English equivalents (`fuentes/05_sintesis_y_lenguaje_2026-10-20/fuentes_cristal_semantico/`). The terms below either follow that glossary or were decided for this translation.

Spanish	English	Note
hilo	thread	The continuity the program follows
reflexión	reflection	The first operation
cerrar, cierre	to close, closure	Four senses, kept apart (Part IX)
cierre propio	own closure	A closure sustained by its own relations
recierre	reclosure	A closure that becomes a unit again
novedad	novelty	A relation that forces the language to be refined
normalización	normalization	Keeping a relation when the reference changes
firma	signature	What an outside observer can read of a unit
indeterminación	indeterminacy	Property of the totality
indefinición	indefiniteness	Not *uncertainty*, which suggests quantum mechanics
singularidad	singularity	In this program: a configuration that defines itself. **Not** the blow-up of a solution of an equation
semilla	seed	What remains in each closure, to emerge again
pared que se corre	moving wall	The mirror that shifts and leaves multiples
recinto	enclosure	A cell bounded by mirrors
sobra / llena justo / no alcanza	exceeds / fills exactly / falls short	How the fractions compare with the totality

Spanish	English	Note
residuo	residue	What a signature does not yet include
falsador	falsifier	A case that refutes a delimited claim

Status marks. The letters keep their Spanish initials: [O] ontology, [D] definition, [T] theorem, [E] reproducible trial (*ensayo*), [F] falsified, [H] physical hypothesis, [A] open (*abierto*).

How to read

What this is. The picture of the program as of today: the ontological thread, the realizations where it is put to the test, and the three branches. Every claim carries its status and its location in this package. This is the reading text of this edition. The previous edition, Imagen Total 3.9, remains complete in `imagen_total_3_9/`, with its full development.

Language guides and supports.

- *Guide:* the thread begins in language, and every topic is presented in the same order: narrative, ontology, language, mathematics, prediction.
- *Support:* within a single argument, each word keeps a single meaning. Part III fixes those meanings and lists the conversions that are not made without proof. Each branch begins with the words that cross from one language into another.

Status.

Mark	What it claims
[O]	Ontology: the author's reading; it is neither proved nor refuted
[D]	A definition or an adopted rule
[T]	Proved under declared hypotheses
[E]	Reproducible trial: the code and the data are in the package
[F]	Falsified: a witness refutes the delimited claim
[H]	Physical hypothesis: the bridge to observables is not yet derived
[A]	Open: an obligation still without proof

In "The thread of two" (*El hilo del dos*), POSTULADO corresponds to [O], DEFINICIÓN to [D] and DERIVADO to [T]. CONTRASTE becomes [H] or [F] depending on its outcome, and ABIERTO becomes [A].

Links. The body of the text does not cite authors. When a result is already known outside the program, it is marked ‹n›, and the links layer at the end (Part X) says where it is. First the thread; then the links.

Reproducing.

- `python3 verificar_integridad.py`, run from the root of the package, checks every file.
- Each realization and each branch has its own verifier, cited where it belongs.

I. Language

1. What is real is what can be described [O]

Human nature is not written only in the genetic code. It also has an immaterial part, which is language. Through selection, language develops a structure that seeks the greatest efficiency in describing reality, and what is real is what can be described.

That structure is inherited, grows and is completed as part of a collective entity. We are the flowers and fruits of an organism whose materiality projects beyond biology and beyond what genetic information contains. Each of us is like one fruit of the tree, or like a mushroom whose spore is the mother structure: an individual projection of a collective knowledge. The difference that gives us identity is the one that keeps novelty alive, so that we can grow beyond the present.

2. The thread [O]

There is a thread. We do not see it whole, but we hold it. It is the continuity between what we distinguished yesterday and what we distinguish today, and that is our construction of reality. The relations in that construction need an ontological representation.

3. To reflect [O]

To reflect is to notice oneself within a larger circumstance: the proprioception of a system inside something that contains it. Reflection does not depend on language; language names it and keeps it. That is why it is the most fundamental layer of the program.

Intuition remains a legitimate way of understanding, because reflection is first seen in oneself.

4. Rule R: a relation exists if it can be named [D]

In the program, a relation exists only if it can be named within the language.

Formally, a language groups realizations into classes, and its reality is the quotient $\Omega/\sim$. A relation ρ is representable in that language if it respects the classes: if $x \sim x'$ and $y \sim y'$, then $\rho(x, y) \Leftrightarrow \rho(x', y')$. If it does not, deciding it requires refining the language: that is a **novelty**.

The same rule reappears in the three branches (Part VI): a question that the previous signature cannot answer, a phase difference that magnitudes do not see, a commutator that the product erases. There it is a consequence of the thread, not its foundation.

5. Counting is the simplest case of language [O · T]

The simplest case of language is counting. The question is which numbers a reference can name when all it has is its first distinction, two, and how that vocabulary grows.

Each prime is a new word. It enters at the step n at which, counting with that prime, doubling n times returns to one for the first time. After the first step, every step brings a new word except one: the 6 ‹3›.

This is “The thread of two”, recovered in full in `fuentes/06_hilo_del_dos/`.

6. The economy of a word [D]

A new word has to do identifiable work: keep an answer, distinguish cases, reduce a declared cost or produce a prediction that can be tested. If a new name changes no operation, it is not progress.

Language can also erase. A formulation that groups as equal some cases that an admitted operation distinguishes loses information; R-02 has an exact witness of this (Part VI, Yang–Mills).

What is resolved does not become unresolved. A new question can separate what an earlier answer grouped together, without making that answer false for its own question.

Every proposal in the program is written in five voices:

1. a faithful narrative;
2. a typed definition;
3. a verifiable proposition;
4. an implementation;
5. a falsifier.

II. The thread [O]

1. The totality. At the beginning there is one: homogeneous, without distinction. Its property is indeterminacy, an open difference with neither magnitude nor direction. The smallest form of that indeterminacy is a bit: plus or minus one, without being able to say which.

2. Reflection. The first operation is a mirror: noticing oneself within something larger. As soon as the unit stops being only one, there are two things at once: what is distinguished, and that from which it is distinguished.

3. The cascade. Reflection repeats in every reflection, and at each step its solutions emerge together, at the same time. Each node reflects the unit again as a ± 1, and around each center N its images $N - 1$ and $N + 1$ appear: around 4, the 3 and the 5; around 8, the 7 and the 9.

Doubling gives depth: 2, 4, 8, 16. Squaring gives reclosure: 2, 4, 16, 256; at each reclosure everything before fits again as a single unit. The 3 is not assumed: it appears with the 4, as the comparisons that two distinctions allow.

4. Normalizing the cardinals. From inside, the cell always measures one. From outside, as the wall moves, each cell is normalized to the totality as a fraction 1/k. When a new multiple appears, a relation that before had nothing to be compared with, everything else is built on it. Two numbers with no common divisor are perpendicular: independent axes, because there is always a remainder.

5. Opening or closing. To close is to have self-referential information: to return to itself after a finite number of steps without losing the relational structure. What decides whether a configuration closes is normalization:

- if the fractions exceed the totality, it closes;
- if they fill it exactly, it repeats;
- if they fall short, it opens.

Angle and identity are the same thing: two mirrors meeting at an angle π/k is the same as applying them k times in a row being doing nothing.

6. Even and odd. What is even closes with its opposite; what is odd stays open.

7. The 6. The 6 is the multiplicity of the 2 and the 3, repeating. A half and a third leave exactly a sixth: the 6 fills the totality without bringing anything new, and that is why the configuration repeats there. In it a closed part, that of the 2, lives together with an open part, that of the 3. It is the same mechanism that makes the 6 the only step of the genealogy without a new word.

8. The 5 and the undefined one. When the one becomes undefined, the golden ratio appears: the number that is one plus its own inverse, the fraction made only of ones that never ends. It is the rectangle which, when a square is removed, leaves another with the same shape as the whole: what remains after distinguishing recognizes itself in the whole.

The golden ratio belongs to the same family as reflection, that of the square minus one. The 5 is a sphere around the 4: it closes only that way, by enveloping, and it cannot repeat on a flat surface.

9. Background and islands. The structure on which everything grows is that of the octaves, that of the indeterminacy of the tetrahedron. The tetrahedron is the background that never closes: each reflection brings something new. Space is in 8 and keeps moving; particles are in 4. On that background, projections appear that do close, such as the cube, and each projection is a configuration that can occur.

10. The singularity. A configuration that defines itself is a singularity: it holds itself up in that space. Its indefiniteness does not surround it and is not a content of it: it is its identity, its own information. Each singularity has its own.

11. The seed. In each closure a seed of the multiplicity remains, with the shape of the whole, as a reference from which to emerge again. Scale remains as the normalizer.

12. The branches. Navier–Stokes, P versus NP and Yang–Mills are three places where this same pair, opening and closing, reappears (Part VI).

Language along the thread [O · editorial reading, for the author's review]

Step	What language gains
1–2	There is something to name, because the unit stopped being only one
3	Vocabulary grows by counting. Each center brings its images, and each reclosure turns everything before into a single word
4	Changing the reference keeps the relation: h/N = hL/NL. The word changes scale without changing meaning
5	What closes can be used again as a unit: an identity that is named
7	The 6 brings no new word
10	Each singularity has a proper name, and its indefiniteness is its information
11	The seed is what is inherited, as language is inherited by each fruit

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III. Words: dictionary and boundaries

This dictionary coordinates the program's uses; it adds no mathematical identities. It brings together the dictionary of types (`fuentes/03_lenguaje/DICCIONARIO_Y_TIPOS.es.md`), the semantic crystal (`fuentes/05_sintesis_y_lenguaje_2026-10-20/05_CRISTAL_SEMANTICO_Y_TIPOS.es.md`) and the refinements of October 24.

Word	In the thread	Formal use	Not to be confused with
Totality, unit	Homogeneous one, without distinction; the reference from which distinctions are made	`1_ref`	A particle, a vertex or a physical bit
Indeterminacy	Property of the totality: an open difference with neither magnitude nor direction	K_Q > 1; Ind	Chance, or probability without an added measure
Bit	Smallest form of indeterminacy: ±1 without being able to say which	A reflection of order two	A bit of physical memory
Reflection, mirror	Noticing oneself within something larger; the first operation	Involution s, with $s^2 = 1$	A material mirror, which is a later laboratory
Difference	A distinction that a rule can certify	Class of ~_Q	Euclidean distance without an added map
Relation	What can be named (rule R)	A relation that respects ~	A binary edge at every level
Novelty	A relation that forces the language to be refined; a new multiple with nothing to be compared with	A question that does not factor through the signature; a new prime, with ord_p(2) = n	A new name that changes no operation
Normalization	From inside, the cell measures one; from outside, 1/k	h/N = hL/NL	Treating the least common multiple, the geometric quotient and the physical period as synonyms
Perpendicular	Two numbers with no common divisor: independent axes	Coprime; counting modulo 6 is counting modulo 2 and modulo 3 ‹2›	Dynamical independence: coprime phases do not prove locking
To close	To have self-referential information that can be normalized	Four senses, kept apart (Part IX)	The complete end of uncertainty
To repeat	The fractions fill the totality exactly	1/p + 1/q + 1/r = 1	To close
To open	The fractions fall short; each reflection brings something new	1/p + 1/q + 1/r < 1; free group	Failure or error
Identity	What a sufficient interface recognizes as a unit	Angle π/k ⇔ (s_i s_j)^k = 1	Equality of inner histories
Singularity	A configuration that defines itself; its indefiniteness is its identity	Typed closure under Q: Ω ≠ ∅ and |Q(Ω)| = 1	The singularity of an equation (blow-up of a solution), a black hole, mass

Word	In the thread	Formal use	Not to be confused with
Seed	What remains of the multiplicity in each closure, to emerge again	—	A new particle
Background	The structure of octaves: the tetrahedron, which never closes	Free product of four Z_2 ‹6›	A pre-existing absolute space
Island, projection	A configuration that does close on the background, such as the cube	Finite reflection group	A fixed physical shape
Moving wall	The mirror that shifts and leaves multiples	CME-002	Physical motion of a wall without a declared law
Residue	What a signature does not yet include	1_res; Ind	Information stored out of necessity
Signature, capsule	A unit with a boundary and with what is needed for the admitted continuations	Γ_G(B) = ∃I R_G(I, B)	A small table in every case
Radius	Declared scale of the reflection	R_i = 1/√(βλ_i)	The proton charge radius, without a bridge
Rhythm, time	Comparison of the rhythms of systems	σ = ℓ/Λ, dimensionless	Absolute seconds without an operational clock
Mass	What is not projected; contained, self-referenced information	$\|V\|^2 + \|m\|^2 = 1$	Kilograms derived from an uncalibrated count
Resolution	What the language already distinguishes; the direction of change: everything returns to one	Res	Entropy as an imported concept
Relational sign	Multiplying or dividing relative to another scale	Exponent ±1	Absolute positive or negative
Tetrahedron	Four registers (one global and three local), or K4, depending on the contract	K4; regular tetrahedron	C4 of four phases, ε = 4
Cube, box	A representation of reflections, axes and accesses	$(Z_2)^3$	Absolute space
Sphere	A node with a perimeter of indefiniteness	—	An elementary particle already identified

Conversions that are not made without proof.

- $\varepsilon = 4$, $q = 4$, $2^2 = 4$, four registers, four phases (C4) and K4 have different types. A numerical coincidence does not make them the same fact.
- Edges are not bits: K4 has six edges, and those are not six independent bits.
- A singularity of the thread is not the blow-up of a solution of Navier–Stokes.
- The arithmetic normalizer is not viscosity, an optical mirror is not a gauge transformation, and semantic uncertainty is not a quantum measurement.
- The commutator is not the spectral gap, and the defect ν is not a mass.
- More images are not more sources, and a recoverable shape does not give access to the same history.
- Recognizing a pattern is not generating it.

IV. Formal core

The minimum that holds the thread up, with its status. The proofs are in the files cited in Part IX.

1. **What is distinguished and what remains undistinguished.** Let C be a set of compatible realizations and Q a family of questions. The realizations fall into classes F_k of weight $\mu_k = |F_k|/|C|$, and then:

   - $\mathrm{Res} = \Sigma\, \mu_k \log_2(1/\mu_k)$ is what the language already distinguishes.
   - $\mathrm{Ind} = \Sigma\, \mu_k \log_2|F_k|$ is what remains undistinguished within each class.
   - **$\mathrm{Res} + \mathrm{Ind} = \log_2|C|$** [T ‹1›]: the totality does not change; its distribution does.

   $K_Q = |C/\sim_Q|$ counts the classes: 0 is inconsistency, 1 is resolution, and more than 1 is sustained indeterminacy [D].

2. **Reflection about a center.** $M_N(d) = (N - d, N + d)$, and $N^2 - (N - d)(N + d) = d^2$ [T]. With $d = 1$, the two images rebuild the center up to one unit: $3 \cdot 5 = 16 - 1$ and $7 \cdot 9 = 64 - 1$.

3. **Genealogy.** The prime p enters at the step n at which doubling n times returns to one modulo p for the first time: $\mathrm{ord}_p(2) = n$.

   - Every n brings a prime of its own, except $n = 1$ and $n = 6$ ‹3›.
   - That is why the center 8 is the only one with no new word above it: $7 \cdot 9 = 63 = 2^6 - 1$, and $9 = 3^2$ is inherited [T].

4. **Reclosure.** The chain 2, 4, 16, 256, 65536 squares at each step. Each step closes the previous one and brings a novelty coprime to everything that was there ‹4› [T].

5. **Novelty, common reference and normalization.** This is R-01 (`../03_RAMAS/yang_mills/R01/`), T1–T4.

   - If the previous signature is x mod N and a prime p appears that does not divide N, the previous signature cannot answer the question modulo p (T1) [T].
   - The common reference $L = Np$ recovers both signatures, and each previous class splits into exactly p new classes (T2 ‹2›) [T].
   - The relation is preserved: $j/N = pj/L$. In general, if h/N passes to a multiple L, $(hL)/(NL) = h/N$, even though the numerator and the representation change (T3) [T].
   - With affine continuations, each part updates from its own residue without knowing the whole: an operational closure (T4) [T].
   - Written falsifiers [F]:
     - a constant map preserves commutation and erases every difference;
     - going from 6 to 18 with the 9 refines without bringing a new prime, because it deepens the 3;
     - the least common multiple gives representability, not dynamics.

- The agreed sequence [D] is: novelty, independent multiple, internal normalization, common reference, own closure, continuity.

6. **Angle and identity.** Two mirrors meet at the angle π/k if and only if (s_i s_j)^k = 1 [T ‹5›]. With three mirrors at angles π/p, π/q and π/r ‹5›:

   - if 1/p + 1/q + 1/r > 1, the enclosure closes;
   - if it equals 1, it repeats;
   - if it is less than 1, it opens.

   CME-002 checks this on 22 enclosures [T · E].

7. **Interface.** Γ_G(B) = ∃I R_G(I, B) replaces the interior by its relation with the boundary B [D]. Adding constraints never enlarges what is compatible: Ω_{n+1} = Ω_n ∩ R_{n+1} [T]. Contextual singularity: Ω ≠ ∅ and |Q(Ω)| = 1, which does not require |Ω| = 1 [D].

8. **Dynamical sufficiency.** A signature Q is sufficient for an evolution F if $Q \circ F = \hat{F} \circ Q$. Across scales, the signature is preserved if $Q = P \circ Q' \circ J$ [D · T in R-02].

9. **Complement.** $\|V\|^2 + \|m\|^2 = 1$ [D]. With $E = \hbar\omega_U$, $pc = \hbar\omega_V$ and $mc^2 = \hbar\omega_m$, this is exactly the relativistic relation between energy, momentum and mass [T].

V. Realizations

These are the places where the thread becomes exact. In them, “to close” means above all that the orientations run out, the first of the four senses (Part IX).

1. The mirror cube

`../02_REALIZACIONES/01_cubo_de_espejos/`

Result	Status	Where
The regular tetrahedron never closes: with at most L reflections it reaches $2 \cdot 3^L - 1$ orientations, and its four mirrors generate a free group ‹6›	[T · E], up to L = 12	`CME-001_cierre_por_reflexion/`
The cube closes at 8 orientations, and the orthoscheme at 48	[T · E]	same
The three inner mirrors of the tetrahedron generate the eight vertices of the cube; parity splits them into two tetrahedra of four	[T]	`geometria_O19_O22/`
22 three-mirror enclosures: it closes if the fractions exceed the totality, repeats if they fill it exactly, and opens if they fall short; angles and identities agree step by step	[T · E]	`CME-002_pared_que_se_corre/`

Result	Status	Where
The bit rule (±1 at each node and exchange of nodes) gives only the angles π/2, π/3 and π/4, and always closes: 48 orientations with three nodes and 384 with four; the 5 needs the golden ratio	[T · E]	`comprobacion_regla_del_bit.py`
Exchanging two of the four references is a reflection. Without a shared reference: π/2; with one shared: π/3; 24 orientations in all. With the bit (the sign of a comparison) π/4 and the cube appear: 48. No angle is put in by hand ‹8›	[T · E]	`CME-003_de_las_cuatro_referencias_a_los_angulos/`
The positions of the images form an infinite orbit; the folded signature brings each image back to the enclosure	[T]	`059_orbita_y_firma_plegada/`
Ideal copy, finite sensor, observational identity, coincidence of returns and optical resonance	[E]	`../04_LABORATORIOS/espejos_E-MIRROR_v1_v8/`

A precision. The criterion by angles π/k is an “if and only if” for cells of Euclidean space whose images fill it without overlapping ‹5›. Counting finitely many orientations is weaker. The triangle with angles 2π/5, 2π/5 and π/5 gives 10 orientations, but its translations shrink by a factor 1/φ without end ‹7›.

CME-003: where π/2 and π/3 come from. They come from the four references of T8, without putting them in by hand [T · E]. Two exchanges that share no reference are independent (π/2). Two that share one run through three references (π/3), and in the cascade 1, 2, 4 that is only possible once the 4 is there.

This is what the author had said [O]: the 2 appears first as a half, with a single mirror; the 3 does not appear until the 4 shows itself, and it emerges as the number of nodes, which are the three comparisons.

That it is impossible to say which of the four is “the one” is what gives the three directions. If one were distinguishable, the enclosure would be flat.

The same tetrahedron gives both:

- reflected in its faces, it creates new references and never closes;
- with the exchange of its references, it closes at 24.

What remains [A]:

- the rule of the moving wall that gives four interchangeable references;
- the selection of the two distinctions (T8);
- the 5, which comes neither from exchanges nor from bits.

2. Spheres, the tetrahedron and four references

`../02_REALIZACIONES/02_esferas/` · `../04_LABORATORIOS/esferas/` · `fuentes/06_hilo_del_dos/v2_8/docs/T8_DEL_LENGUAJE_AL_ESPACIO_v2_8.md`

- Each sphere is a node with a perimeter of indefiniteness [O].
- Three centers fix a plane and leave two reflected extensions, above and below [T · E, model v10].
- Four equal, mutually tangent spheres have six contacts, and their centers form a regular tetrahedron [T].
- The 3 is the $2^2 - 1$ comparisons of two distinctions, and the four references realize the 24 permutations [T].
- The fifth sphere touching four equal ones has radius $(\sqrt{6} - 2)/2$: the fifth fits only with another radius [T]. Reading it as the event of a new distinction is [O · H].
- **The face mirror** creates a fifth reference at 5/3 of the radius. With the four faces, the next tetrahedral layer forms, with a fixed focus (O20) [T].
  - That point is the center of the opposite face of the neighboring octahedron.
  - Reflecting and filling are different operations (O22) [T].
  - The factor 5/3 depends on choosing the face mirror; it is not a physical law.
  - It is in `../02_REALIZACIONES/01_cubo_de_espejos/geometria_O19_O22/`.
- **Trajectories [A].** A trajectory that runs along a curvature can continue on the neighboring sphere with the opposite turn. This is a proposal of the laboratory. It keeps three models apart:
  - the contact graph;
  - ideal tangential transport;
  - the optics of real glass.

3. The C4 circuit

`../02_REALIZACIONES/03_circuito_C4/` · `../05_ARCHIVO/navier_stokes_O54_O92/O72_…` and `O74_…`

- Four reflections give 16 states. Reflecting each vertex of the tetrahedron in its opposite face gives a new regular tetrahedron, at scale 5/3 [T].
- With ideal contacts, s_i + s_j = 0 on each link. The ring of four admits an alternating turn (a, −a, a, −a), and the triangle and K4 do not: dim ker C4 = 1 and dim ker C3 = dim ker K4 = 0 [T].
  - The flexing of the quadrilateral and that turn are different freedoms.
- With different arms, the turn returns to itself if H = ∏ a_i/b_i = 1. With the radii as arms, the product cancels in pairs and equals 1 for any collection of radii [T].
  - For the turn to return in time, the phases must also be compatible [A].
- **Next trial [A].** Two configurations with the same C4 graph and a different transmission law. Check that only one sustains the chosen return, and repeat with K4.

VI. Branches

The three branches are places where the pair of opening and closing reappears. The grammar they share has three features, and it comes from the thread; it does not replace it:

- novelty is a question that the previous signature cannot answer;
- closure is a sufficient signature;
- loss happens by blocking.

No branch solves the Millennium Prize Problem it corresponds to. Each has exact results within its contract, and a precise limit.

1. Navier–Stokes

`../03_RAMAS/navier_stokes/` · current: `VIGENTE_O93/` (October 22) · notes: `revision_2026-10-24/`

Words that cross.

- *Singularity* in the thread is a closure that holds itself up. In the equation it is something else: the solution ceasing to be smooth. The two are not interchanged.
- *To normalize* is to divide by the scale of the flow itself.
- *Seed* [O]: in each closure a seed of the multiplicity remains, and scale remains as the normalizer.

Contract. A viscous, incompressible fluid on the three-dimensional torus, with no external force ‹NS1›.

What holds.

- Double normalization: $W = M^2D$, $B = M^2D_2$ and $P = M^3\mathcal{T}$ [T]. The last is a scaling identity, because the term that stretches vorticity is cubic.
- Exact values for a trigonometric field: $W = 95/9$, $B = 266/9$ and $P = \pm 3/4$ [T · E]. They were confirmed another way, in physical space.
- A certificate for a subclass ‹NS4›: $L = 6$, $M = 9$, $\mathfrak{M}_0 \approx 0.5195 > 1/2$ and $\mathfrak{M}_1 \le 0.2717 < 1/2$ [T · E].
- With the same spectrum of magnitudes, phases can reverse the sign of the stretching [T · E].
- O93 and the verifiers of O80–O86 and O92 reproduce bit for bit [E].

The limit.

- The O93 criterion, $\int_0^T [M\mathcal{T} - \nu D_2]_+/D \, dt < \infty$, is sufficient to continue the solution. But its integrand is $\tfrac{1}{2}[(\log W)']_+$, and the integral is finite if and only if the solution does not blow up. It is an exact restatement of the obstacle, not a bound [T, conditional contract].
- E4 only checks the sign of a formula. E6 runs to $t = 0.08$, too short to say anything about regularity.
- Falsifiers that are part of the state of the branch:
  - O56: the radii and the arithmetic normalizer do not decide whether a material perpendicularity is preserved; the cross strain $a^T S b$ is needed;
  - O75: when the four-link reference is transported by the flow, a contribution appears that four nodes alone do not determine;
  - O76: because pressure is non-local, the positions of four capsules and their velocity derivatives up to order two are not enough to predict their accelerations;
  - O77: in a family of smooth rings, the evolution produces terms that five parameters cannot rebuild; that finite seed is not enough, although this does not prove that none is;
  - O85: the certification margin of O84 does not change under exact critical scaling, so concentrating the flow does not improve it;
  - O86: magnitudes are not enough, because phases change the sign of the stretching;
  - O88: phase closure, four-link closure, periodic return, identity and regularity are five different claims;
  - O90: the common 2:1 locking can be lost under arbitrarily small perturbations without breaking the smooth evolution: losing synchrony does not imply blow-up.

The program's own route. `modelo_propio_018/` is the branch's only path toward a theorem about an equation of its own [A].

- The per-octave gains g_j have to be derived from the mirror rule, and only then compared with the known threshold ‹NS3›.
- It is enough to have $\gamma_j \gtrsim 2^{j/2}$. With a logarithmic margin, the condition is $\Sigma\,(1 + \gamma_j)/2^{j/2} = \infty$.
- The accumulated product of the gains must be controlled, not only their limit $\sqrt{2}$.
- What results is a new system, not Navier–Stokes; it stands as a conditional program for a theorem.

Open [A]. Bound the global sum of defects and reference jumps without knowing the future of the flow in advance.

2. P versus NP

`../03_RAMAS/p_vs_np/` · current: `VIGENTE_LAB_CONV_V2/` (October 22) · the program's own bound: `fuentes/06_hilo_del_dos/v2_8/docs/PISO_DEL_CAMINO_v2_7.md` (October 14)

Words that cross.

- *Closure*: a sufficient boundary interface, Γ_G(B).
- *Novelty*: a question that the signature cannot answer.
- *Count and radius are not a signature* [F]. A = {a ≠ b} and D = {b ≠ c} have the same 18 assignments and the same radius, and answer differently to a new relation.

What holds [T · E].

- Exact normalization of three-coloring interfaces by renaming colors ‹P5›.
- Certified closure by the diamond.
- Odd-cycle obstruction in a neighborhood.
- Comparison of elimination orders.
- Exhaustive verification on 1,338 small graphs, and agreement with an independent solver on 300 random graphs.

The barrier of the method: N1 [T].

- *N1-A.* No order brings the table engine below 3^(tw+1), where tw is the treewidth of the graph ‹P1›.
- *N1-B*. With pruning, every run that concludes "there is no coloring" is a resolution refutation. That is why it costs 2^Ω(n) on random graphs, with any selector, even with an oracle ‹P2›.

The measurement confirms it: cost follows width, not difficulty [E].

- On k × k grids, which are trivial, evaluations go from 22 to 107,005 between k = 3 and k = 9.
- On random 3-regular graphs, from 105 to 178,406 between 16 and 56 vertices.

The limit.

- N1 is the barrier of the table engine; it does not decide P versus NP. Under the SETH hypothesis, no exact refinement of those tables lowers the base 3 ‹P3›.
- Merging two vertices that are forced to share a color is not free. Recognizing that case is coNP-complete ‹P4›, and the claim is vacuous if the graph has no coloring.

Open [A].

- An operation expressively different from tables. Before measuring its cost, one must prove that it escapes simulation by resolution.
- Section VII of LAB-CONV v2 should cite N1.

3. Yang–Mills

`../03_RAMAS/yang_mills/` · current: `VIGENTE_R02/` (October 23) · predecessor: `R01/`

Words that cross.

- *Loop*: what remains of the reference after going around.
- *Signature*: (x, y, z) = (tr U/2, tr V/2, tr UV/2).
- *Blocking*: (U, V) → UV.
- The *gap*, the commutator and ν are three different things.

What holds [T · E].

- Gauge transport of loops and conservative refinement.
- For a pair of anchored SU(2) loops, the signature (x, y, z) is complete up to simultaneous conjugation ‹Y1›. The converse direction was checked on 2,000 pairs, in both orientations, with error ≤ $1.5 \cdot 10^{-13}$.

- **Blocking erases [F].** (U, V) → UV preserves the product and erases the commutator. It is the exact witness that language can lose a difference (Part I, §6).
- Dynamical sufficiency $Q \circ F = \hat{F} \circ Q$, and preservation across scales $Q = P \circ Q' \circ J$.

The triangle [T].

- The central quantity, $1 - x^2 - y^2 - z^2 + 2xyz$, is the Gram determinant of three unit vectors: it is the trigonometry of a spherical triangle.
- That it is never negative is the triangle inequality.
- The defect $\nu = 2(1 - x^2 - y^2 - z^2 + 2xyz)$ vanishes when the triangle flattens, and that happens exactly when the two loops commute. It is the same sector as that of the orthogonal mirrors of the cube, which commute.

This agrees with the author's reading [O]: the basis of equilibrium is a trigonometric equation, at bottom a theorem about triangles.

The limit. Sufficiency for two loops is not inherited by a large lattice ‹Y2›, and no result of the branch is a gap.

Open [A].

- R-03: two plaquettes, the lattice action, the blocking a → 2a and an SU(2) versus U(1) control. Before opening it, it has to be placed against the block renormalization that is already rigorous ‹Y3›.
- The underlying obligation still stands: Yang–Mills in four dimensions with a uniform positive gap.

VII. Mass, time and gravity

The reading [O]

Mass is the complementary part of the projection: what the image holds up against the others, what is not projected. It is the mechanism by which information is stored, and the ratio between two radii is the point.

Time is the comparison of the rhythms at which systems happen. It is like a musical measure that each instrument subdivides and fills. Atoms and particles give the original rhythm, and the other rhythms are multiples or harmonics of that projection.

Comparison has no direction; direction lies in resolution. As the coherence of space, understood as storage of information, is resolved, everything returns to one.

Only the present exists, as the geometric storage of information; the past is inherited information. Our time is finite in one, like a single cycle; the rest of time is comparative.

Physical promise: what is exact and what law is missing

Idea	What is exact	The missing law	Test	Status
Mass as the complement of the projection	$\|V\|^2 + \|m\|^2 = 1$ is the relativistic relation between energy, momentum and mass [T]	The rule that fixes the scale of each image	With a single scale for all particles, it fails for the leptons: the depths of the electron (23.53) and the muon (8.14) are not integers [F]	Complement [T]; single scale [F]; rule [A]
Mass as contained, self-referenced information	A normalized content equals one in many different states [T]	A quantity that changes between states and has units of mass	—	[O · H]
Radius and rhythm	$R_i \omega_i = \sqrt{(\Gamma/\beta)}$ in the comparator model; with the spectral convention $R = 1/\sqrt{\lambda}$, the product is $\sqrt{\Gamma}$. With K4: $R = a/(2\sqrt{\beta})$ and $\omega = 2\sqrt{\Gamma}/a$ [T]	The program's rule that fixes a, β and Γ	Replacing a by sa keeps the closure and scales the radius: connectivity alone does not fix a length [T]	[T] in the model; [A] the rule
Proton	Anchor A: $4\hbar/(m_p c) \simeq 0.841235640$ fm; anchor B: $\sqrt{15}\cdot\hbar/(m_p c) \simeq 0.814522906$ fm [E, within the model]	The charge density and its form factor	Test against a datum that was not used to fix the choices; do not choose the anchor after seeing the measured value	[H]
Flywheel	$C_\delta = 4J \sin^2(\delta/2)$, and for small turns $C_\delta \approx J\delta^2$ [T]	An action, or a response to an impulse, that makes J an inertia	—	[H]
Electron and proton	Refinement of October 23: every structure has structural self-reference; what distinguishes them is their degree of definition relative to the space of uncertainty [O]	Observables and an independent prediction	—	[H]

Idea	What is exact	The missing law	Test	Status
Time	Moving frame x = X(t) + ℓ(t)A(t)y; three phases give two independent differences. The ratio Γ = κ_i/κ_0 does not change when the phase zero is moved (O45). A discrete clock with no external clock: Ω = 2 arcsin(√(ηλ)/2), and dispersion-free propagation fixes η = 1 (MF-43) [T, conditional]	A synchronization law with a shared reference; a step is not a second	Fluid clocks are one realization [E]	[O · H]
Gravity	The Pythagorean decomposition, in whatever space has been defined [T]. Global closure without equal labels: 8 coherent states out of 64 on K4 (GR3) [T]	A dynamical law, a metric or field, and the Newtonian and relativistic limits	A pure phase-mismatch potential, K(r)(1 − cos δ), does not attract at synchrony (October 8) [F, for that candidate]. The gravitational repulsion of antimatter was falsified and withdrawn ‹F1› [F]	[H]
Neutron	—	The author's proposal: its lifetime, through an equivalent radius referred to an atomic radius, estimating the collapse from the same positions	—	[A]

Outside the core. Dark matter as the complementary fraction 1 − R was exploratory. It is not part of the program.

Where.

- `imagen_total_3_9/Imagen_Total_3_9_Informe_General.md`, chapter 9.
- `../02_REALIZACIONES/01_cubo_de_espejos/Revision_Radio_Dinamico_Cubo_2026-10-23.md`.
- `fuentes/06_hilo_del_dos/v2_8/docs/MASA_COMO_COMPLEMENTO_v2_6.md`.

- `fuentes/04_sintesis_integral_2026-10-20/06_GEOMETRIA_MASA_GRAVEDAD_ESPEJOS.es.md`.
- `SUPLEMENTOS/Suplemento_Ritmo_Ciclo_Base_Tiempo_y_Gravedad_2026-10-24.en.md`: base cycle, MF-43, O45, GR3 and the gravity falsifier.
- `fuentes/07_ritmo_y_gravedad_recuperados/`: GR3 and the falsifier of October 8, recovered byte for byte.

VIII. Language at work

The semantic crystal. It organizes each term with its history, etymology, uses, symbols and translation, without turning every association into an equality. The pair *casa* (house) and *hogar* (home) is the minimal test: two words can share the building and still distinguish belonging. It is the support of Part III. It is in `fuentes/05_sintesis_y_lenguaje_2026-10-20/fuentes_cristal_semantico/`.

ROSI. It records questions, obligations, witnesses, sources, costs, falsifiers and coverage. A route that skips an obligation cannot win by looking faster. SHA-256 fingerprints identify files, not reviewers. It is described in chapter 10 of edition 3.9.

Emergence laboratories. They are in chapter 10 of edition 3.9.

- *Chemistry.* The descriptor r_V = 2/(g − 10) lowers the error from 53.520874 to 50.797514, about 5.09%. But an equivalent transformation of the group number reproduces the same information: it is useful in that trial and adds no independent information [E · F]. For the transition metals, a quadratic reference in the group number can beat the descriptor [F].
- *Growth.* 80 trajectories in 40 pairs. The Jaccard value 0.976287 is listed as reported, but it cannot be rebuilt from the available microstates [E, partial].
- *Quantum control.* For two independent qubits, the probability of projecting onto the triplet is (3 + n·m)/4 [T].

Communication. The infinite monkey and the essay “I am not a robot yet, I have the right to a tantrum” (*Aún no soy un robot, tengo derecho a un berrinche*) test which relations a reader manages to rebuild. They accompany the program; they are not proofs.

IX. Matrix

What holds each step of the thread

Step	What is checked	Status	Where
1 · totality and bit	Definition	[O]	`fuentes/06_hilo_del_dos/v2_8/docs/EL_BIT_ABIERTO_v2_3.md`
2–3 · reflection ±1	$N^2 - (N-1)(N+1) = 1$	[T]	`HILO_DEL_DOS_FORMALIZACION_v2.md`, §5
3 · genealogy and reclosure	3 and 5 new at the 4; 7 new and 9 inherited at the 8 ‹3›; chain 2, 4, 16, 256 ‹4›	[T]	same, §7–§9
3 · the 3 with the 4	The 3 is the $2^2 - 1$ comparisons of two distinctions; the four references realize the 24 permutations	[T]	`DEL_CUATRO_A_LA_DISTANCIA_Y_EL_TIEMPO_v2_7.md`, T8

Step	What is checked	Status	Where
3–5 · the angles π/2 and π/3	Exchanging references is reflecting: with no shared reference, π/2; with one shared, π/3; with the bit, π/4. In all 24 chambers the enclosure is the same	[T · E]; the map from the wall is [A]	`../02_REALIZACIONES/01_cubo_de_espejos/CME-003_de_las_cuatro_referencias_a_los_angulos/`
4 · coprime and perpendicular	Counting modulo 6 is counting modulo 2 and modulo 3 at once ‹2›	[T]; the reading “perpendicular” is [O]	O28, R-01
5 · exceeds, fills, falls short	22 three-mirror enclosures; angles and identities agree step by step	[T · E]	`../02_REALIZACIONES/01_cubo_de_espejos/CME-002_pared_que_se_corre/`
5 · the bit rule	Only π/2, π/3 and π/4; it always closes (48 with three nodes, 384 with four)	[T · E]	`comprobacion_regla_del_bit.py`, same folder
6 · even and odd	Two mirrors at π/k contain the half turn, which takes each thing to its opposite, if and only if k is even	[T]	—
7 · the 6	Exact supports: 1/2 + 1/3 + 1/6 = 1; the enclosure of the 6 is that of the 3 times that of the 2 (12 = 6 · 2); in the genealogy, the 6 brings no new prime, because its own part of $2^6 - 1$ equals 3	[O] reading stated by the author: the same mechanism, the multiplicity of the 2 and the 3 repeating. [T] the two facts, each in its own domain. [A] a map between the two realizations that preserves their operations; the numerical equality alone does not give it	CME-002; `HILO_DEL_DOS_FORMALIZACION_v2.md`, §8
8 · the 5	$\varphi^2 - 1 = \varphi$ and its continued fraction is all ones; the angle π/5 uses φ; in the plane its translations shrink by 1/φ without end ‹7›, and it closes only on the sphere (icosahedron, 120 orientations)	Supports [T]; “the undefined one” is [O]; the step of the cascade where it appears is missing [A]	CME-002
8 · the fifth sphere	Radius $(\sqrt{6} - 2)/2$, and $\sqrt{6} = \sqrt{(5^2 - 1)/2}$	[H]: the same family of the square minus one, without derivation	T8 (c)

Step	What is checked	Status	Where
9 · background and islands	The regular tetrahedron never closes ‹6›; the cube closes	[T · E]	CME-001
10 · singularity	Every enclosure that closes has a point that all its mirrors leave fixed; the singularity's own indefiniteness is the Ind of Res + Ind = $\log_2\|C\|$	[T] the fixed point and the measure; [O] identity = indefiniteness	`HILO_DEL_DOS_FORMALIZACION_v2.md`, §4
11 · seed	—	[O · H]	—

Four senses of "to close" that must not be mixed

1. **Orientations:** the number of distinct turns and reflections runs out (CME-001, CME-002).
2. **Observation:** a signature is enough for the declared questions.
3. **Dynamics:** the signature can update itself under the evolution, $Q \circ F = \hat{F} \circ Q$.
4. **Positions:** the images of a point. In the cube they form an infinite orbit, which the folded signature of O59 brings back to the enclosure.

What is open, in order

In the thread

1. The rule of the moving wall that gives four interchangeable references. This is what remains of CME-003, which already derives π/2 and π/3 from those references.
2. The formal bridge of the 6 between the genealogy and the mirrors.
3. The step of the cascade where the one becomes undefined and the 5 appears.
4. How the genealogy (new or inherited) relates to normalization (closes, repeats or opens). The 4 is inherited and closes; the 9 is inherited and opens; the 7 is new and opens.

In the realizations

5. The law of tangential trajectory between spheres, and the C4 trial with two transmission laws.

In the branches

6. Navier–Stokes: the per-octave gains, derived from the mirror (O18).
7. P versus NP: an operation outside tables that escapes resolution.
8. Yang–Mills: R-03, placed against block renormalization.

In physics

9. The scale rule for mass.
10. The law of synchronization of rhythms.
11. The neutron.

In the editing

12. The pending items of each `revision_2026-10-24/`:
    - Navier–Stokes: state in O93 what its criterion is, and bring the citations to it;
    - P versus NP: have LAB-CONV v2 cite N1;
    - Yang–Mills: have R-02 name the classical theorem.

Control criterion. If the thread is on the right path, it closes by itself. Wherever a closure has to be forced, that is the signal for revision.

X. Links layer

What others have already found, and where the thread meets it. It is not the foundation of the program: it serves to place each result and check its reach.

Thread and core

- ‹1› C. E. Shannon, "A Mathematical Theory of Communication", *Bell System Technical Journal* 27 (1948). Res + Ind = $\log_2|C|$ is the chain rule.
- ‹2› The Chinese remainder theorem (classical).
- ‹3› K. Zsigmondy, "Zur Theorie der Potenzreste", *Monatshefte für Mathematik und Physik* 3 (1892). The only exceptions are n = 1 and n = 6.
- ‹4› The Fermat numbers are pairwise coprime (Ch. Goldbach, letter to L. Euler, 1730).
- ‹5› H. S. M. Coxeter, "Discrete groups generated by reflections", *Annals of Mathematics* 35 (1934); "The complete enumeration of finite groups of the form R_i² = (R_iR_j)^{k_ij} = 1", *Journal of the London Mathematical Society* 10 (1935).
- ‹6› S. Świerczkowski, "On chains of regular tetrahedra", *Colloquium Mathematicum* 7 (1959). M. Elgersma and S. Wagon, "The Quadrahelix: A Nearly Perfect Loop of Tetrahedra", arXiv:1610.00280.
- ‹7› The crystallographic restriction: in the plane there is no translational symmetry of order 5 (classical).
- ‹8› E. H. Moore, "Concerning the abstract groups of order k! and ½k! holohedrically and hemihedrically isomorphic with the symmetric and the alternating substitution-groups on k letters", *Proceedings of the London Mathematical Society* 28 (1897). The exchanges of n things form a reflection group; CME-003 uses this for four references.
- CME-002 also uses the growth series of R. Steinberg, *Endomorphisms of Linear Algebraic Groups*, Memoirs of the AMS 80 (1968), and D. H. Lehmer's number for (2, 3, 7): "Factorization of certain cyclotomic functions", *Annals of Mathematics* 34 (1933).

Navier–Stokes

- ‹NS1› Clay Mathematics Institute, official problem statement: https://www.claymath.org/millennium/navier-stokes-equation/. On September 8, 2026, OpenAI announced the formation of a singularity with a smooth external force, presented as alternatives C and D: https://openai.com/index/navier-stokes-solution/. On September 11, Clay published a statement: https://www.claymath.org/news/navier-stokes-announcement/. That announcement does not answer the force-free question studied in this branch. Sources checked on October 23 for edition 3.9.
- ‹NS2› Z. Lei and F. Lin, "Global mild solutions of Navier–Stokes equations", *Communications on Pure and Applied Mathematics* 64 (2011). It is already cited in O80 and O81, together with Pizzocchero and Chemin–Gallagher–Paicu; it still has to be carried into O93.
- ‹NS3› J.-L. Lions, *Quelques méthodes de résolution des problèmes aux limites non linéaires* (1969). T. Tao, "Global regularity for a logarithmically supercritical hyperdissipative Navier–Stokes equation", *Analysis & PDE* 2 (2009). D. Barbato, F. Morandin and M. Romito, "Global regularity for a slightly supercritical hyperdissipative Navier–Stokes system", *Analysis & PDE* 7 (2014).
- ‹NS4› S. Chernyshenko, P. Constantin, J. Robinson and E. Titi, "A posteriori regularity of the three-dimensional Navier–Stokes equations from numerical computations", *Journal of Mathematical Physics* 48 (2007). It is the direct precedent of the certificates of O82–O84.

P versus NP

- ‹P1› R. Dechter, "Bucket elimination: a unifying framework for reasoning", *Artificial Intelligence* 113 (1999): variable elimination and treewidth.
- ‹P2› P. Beame, J. Culberson, D. Mitchell and C. Moore, "The resolution complexity of random graph k-colorability", *Discrete Applied Mathematics* 153 (2005).

- ‹P3› D. Lokshtanov, D. Marx and S. Saurabh, “Known algorithms on graphs of bounded treewidth are probably optimal”, SODA 2011. Under the SETH hypothesis, no exact refinement of boundary tables lowers the base 3.
- ‹P4› P. Kilby, J. Slaney, S. Thiébaux and T. Walsh, “Backbones and backdoors in satisfiability”, AAAI 2005.
- ‹P5› A. Nerode, “Linear automaton transformations”, *Proceedings of the AMS* 9 (1958): the Myhill–Nerode congruence.
- Clay Mathematics Institute: https://www.claymath.org/millennium/p-vs-np/.

Yang–Mills

- ‹Y1› H. Vogt, “Sur les invariants fondamentaux des équations différentielles linéaires du second ordre”, *Annales scientifiques de l’École Normale Supérieure* 6 (1889), and R. Fricke (1896). T3 and the formula for ν are that theorem. The ontological reading and the assembly of falsifiers are the program’s own.
- ‹Y2› C. Procesi, “The invariant theory of n × n matrices”, *Advances in Mathematics* 19 (1976). R. Giles, “Reconstruction of gauge potentials from Wilson loops”, *Physical Review D* 24 (1981).
- ‹Y3› T. Bałaban, series on the renormalization group in lattice gauge theories, *Communications in Mathematical Physics* (1985–1989).
- Clay Mathematics Institute: https://www.claymath.org/millennium/yang-mills-the-maths-gap/.

Physics

- ‹F1› ALPHA Collaboration, “Observation of the effect of gravity on the motion of antimatter”, *Nature* 621 (2023).

Ontological positioning (being built by the author)

- C. Rovelli, “Relational quantum mechanics”, *International Journal of Theoretical Physics* 35 (1996).
- J. A. Wheeler, “Information, physics, quantum: the search for links” (1989).
- J. Barbour, *The End of Time* (1999).

Provenance

- **This package.** `00_LEER_PRIMERO.md` (in English, `00_READ_FIRST.en.md`) says what is current, what was recovered and what was withdrawn. `MANIFIESTO_SHA256.txt` and `verificar_integridad.py` check every file. `PROCEDENCIA.csv` says which package each one came from.
- **Sources of the thread.** `fuentes/`: foundations, formal core, language, the synthesis of October 20, the semantic crystal and “The thread of two” in full, with 37 documents of October 13 and 14 recovered byte for byte.
- **Agreed outline.** `ESQUEMA_ACORDADO_2026-10-24.md`: the order, the author’s refinements and where each idea already was.
- **Previous edition.** `imagen_total_3_9/`: Imagen Total 3.9, of October 23.

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